

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

EXPERIMENT NO.: 02

Design Webpage using HTML5 & CSS3

Student Name: Ch. Venu Gopal Reddy

Roll No.: A24126552078

College: Anil Neerukonda Institute of Technology and Sciences (ANITS)

Course Outcome: CO1

Date:

1. TITLE

Design Webpage using Basic and Medium CSS Concepts

2. AIM

To design and style a webpage using basic and medium CSS concepts with an external stylesheet.

3. OBJECTIVE

To understand and implement CSS selectors, colors, typography, navigation styling, Flexbox, box model properties, pseudo-classes, tables, forms, transitions, transforms, and external CSS.

4. THEORY

Cascading Style Sheets (CSS) is used to control the presentation and visual appearance of HTML webpages. CSS separates the structure of a webpage from its design and allows common styling rules to be applied consistently.

This experiment demonstrates basic CSS features such as universal and element selectors, colors, spacing, borders, typography, and navigation styling. It also demonstrates medium-level concepts including Flexbox, class selectors, box shadows, transitions, transforms, pseudo-classes, table styling, form controls, and focus effects.

The webpage uses an external CSS file named style.css, connected to index.html using the <link> element.

5. ALGORITHM / PROCEDURE:

1. Create index.html and define the basic HTML document structure.
2. Add the header, navigation bar, main sections, skills cards, student table, contact form, and footer.
3. Create the external stylesheet style.css.
4. Apply selectors, colors, fonts, spacing, borders, and layout properties.
5. Use Flexbox to arrange the skill cards.
6. Apply hover, focus, transition, transform, table, and form styling.
7. Save both files in the same folder and link style.css from index.html.
8. Open index.html in a browser and verify the webpage and interactions.

6. HOW TO RUN THE PROGRAM

- Place index.html and style.css in the same folder.
- Open the folder in VS Code.
- Open index.html using Live Server or directly in a browser.
- Verify the styling, table, form fields, and hover/focus effects.

7. PROGRAM:

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
```

```
<title>CSS Demonstration</title>

<!-- External CSS -->
<link rel="stylesheet" href="style.css">
</head>

<body>

  <header>
    <h1>CSS Demonstration</h1>
    <p>Basic and Medium CSS Concepts</p>
  </header>

  <nav>
    <a href="#">Home</a>
    <a href="#">About</a>
    <a href="#">Courses</a>
    <a href="#">Contact</a>
  </nav>

  <main>

    <section>
      <h2>About CSS</h2>

      <p>
        CSS stands for Cascading Style Sheets.
        It is used to style and design HTML webpages.
      </p>
    </section>

    <section>
      <h2>My Skills</h2>

      <div class="cards">

        <div class="card">
          <h3>HTML</h3>
          <p>Used to create the structure of a webpage.</p>
        </div>
        <div class="card">
          <h3>CSS</h3>
          <p>Used to style and design the webpage.</p>
        </div>
        <div class="card">
          <h3>JavaScript</h3>
          <p>Used to add functionality and interaction.</p>
        </div>

      </div>
    </section>

    <section>
      <h2>Student Details</h2>
```

```

        <table>
            <tr>
                <th>Name</th>
                <th>Branch</th>
                <th>Year</th>
            </tr>

            <tr>
                <td>Venu Gopal</td>
                <td>Computer Engineering</td>
                <td>2nd Year</td>
            </tr>

            <tr>
                <td>Rahul</td>
                <td>Information Technology</td>
                <td>2nd Year</td>
            </tr>
        </table>
    </section>

    <section>
        <h2>Contact</h2>

        <form>
            <input type="text" placeholder="Enter your name">
            <input type="email" placeholder="Enter your email">

            <textarea placeholder="Enter your message"></textarea>

            <button type="submit">Submit</button>
        </form>
    </section>

</main>

<footer>
    <p>© 2026 CSS Demonstration</p>
</footer>

</body>

</html>

```

style.css (External CSS)

```

/* ===== BASIC CSS ===== */

/* Universal Selector */
* {
    box-sizing: border-box;
}

/* Element Selector */

```

```
body {
    margin: 0;
    font-family: Arial, sans-serif;
    background-color: #f2f2f2;
    color: #333;
}

/* Header */
header {
    background-color: #2563eb;
    color: white;
    text-align: center;
    padding: 30px;
}

header h1 {
    margin: 0;
    font-size: 36px;
}

header p {
    font-size: 18px;
}

/* Navigation */
nav {
    background-color: #222;
    text-align: center;
    padding: 15px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 0 20px;
}

/* Pseudo-class */
nav a:hover {
    color: #60a5fa;
}

/* Main */
main {
    width: 900px;
    margin: 30px auto;
}

/* Sections */
section {
    background-color: white;
    margin-bottom: 25px;
    padding: 25px;
    border: 1px solid #ddd;
    border-radius: 10px;
}
```

```

}

h2 {
  color: #2563eb;
  border-bottom: 2px solid #2563eb;
  padding-bottom: 8px;
}

/*===== MEDIUM CSS ===== */

/* FLEXBOX */
.cards {
  display: flex;
  gap: 20px;
}

/* Class Selector */
.card {
  flex: 1;
  background-color: #eff6ff;
  padding: 20px;
  border-radius: 10px;

  /* Box Shadow */
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
  /* Transition */
  transition: 0.3s;
}

/* Hover + Transform */
.card:hover {
  transform: translateY(-5px);
  box-shadow: 0 8px 18px rgba(0, 0, 0, 0.2);
}
.card h3 {
  color: #1d4ed8;
}

/* Table */
table {
  width: 100%;
  border-collapse: collapse;
}

th,
td {
  border: 1px solid #ccc;
  padding: 12px;
  text-align: left;
}

th {
  background-color: #2563eb;
  color: white;
}

```

```

/* Pseudo-class */
tr:nth-child(even) {
    background-color: #f1f5f9;
}

/* Form */
input,
textarea {
    width: 100%;
    padding: 12px;
    margin-bottom: 15px;

    border: 1px solid #ccc;
    border-radius: 6px;
}

/* Focus Pseudo-class */
input:focus,
textarea:focus {
    border-color: #2563eb;
    outline: none;
}

/* Button */
button {
    background-color: #2563eb;
    color: white;

    padding: 12px 25px;

    border: none;
    border-radius: 6px;

    cursor: pointer;

    transition: 0.3s;
}

button:hover {
    background-color: #1d4ed8;
    transform: scale(1.05);
}

/* Footer */
footer {
    background-color: #222;
    color: white;
    text-align: center;
    padding: 20px;
}

```

8. CODE EXPLANATION

External CSS: The `<link>` element connects `index.html` with `style.css`.

Selectors: Universal, element, class, and pseudo-class selectors target different webpage elements.

Layout: The main container is centered, and Flexbox arranges the skill cards.

Styling: Colors, spacing, borders, border-radius, typography, and box shadows improve presentation.

Interactions: Hover and focus pseudo-classes provide visual feedback, while transition and transform create effects.

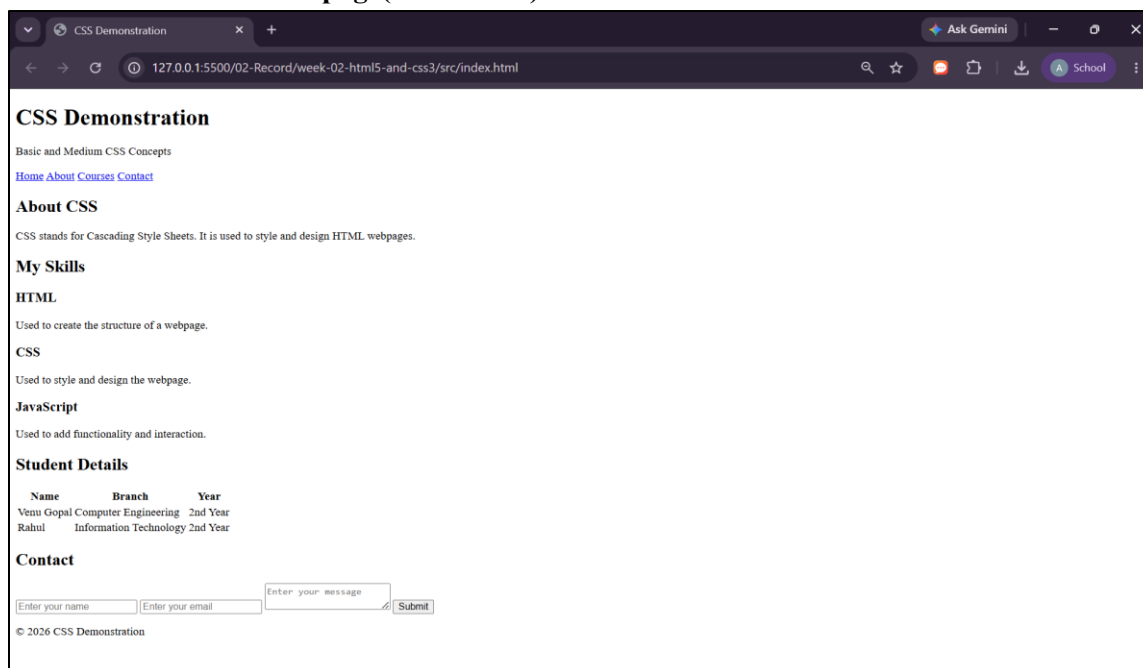
Table and Form: The table uses borders and alternate rows, while form controls receive consistent spacing and focus styling.

9. TEST CASES AND EXECUTION DATA

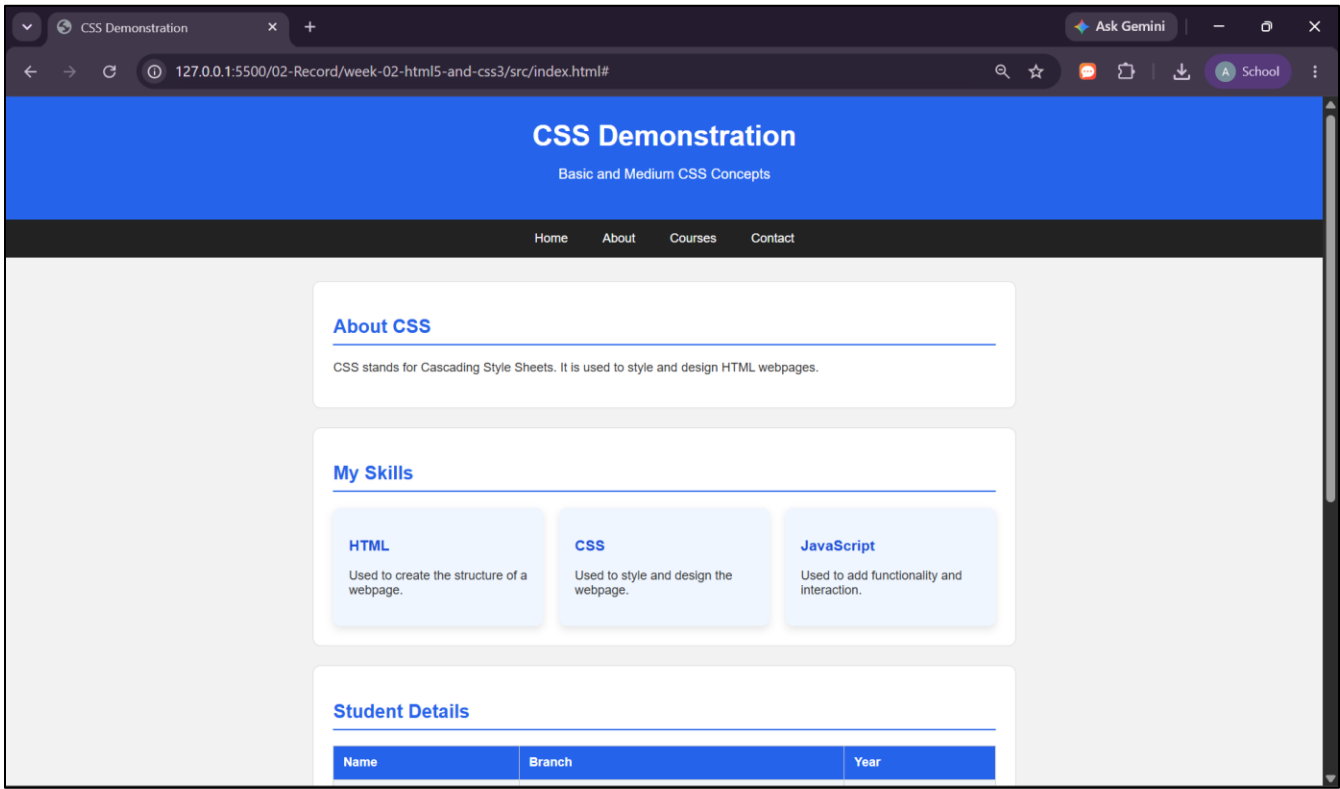
Test Case	Input / Action	Expected Result	Actual Result	Passed
TC01	Open index.html	Webpage loads with CSS styling	Webpage loads successfully	Yes
TC02	Hover over navigation link	Link color changes	Hover effect works	Yes
TC03	Hover over a skill card	Card moves upward and shadow changes	Transform and shadow work	Yes
TC04	Enter data in form fields	Fields accept input and show focus styling	Input and focus styling work	Yes
TC05	Hover over Submit button	Button changes appearance and scales	Button hover effect works	Yes
TC06	View student table	Borders, header, and alternate rows are visible	Table styling works	Yes

10. OUTPUT

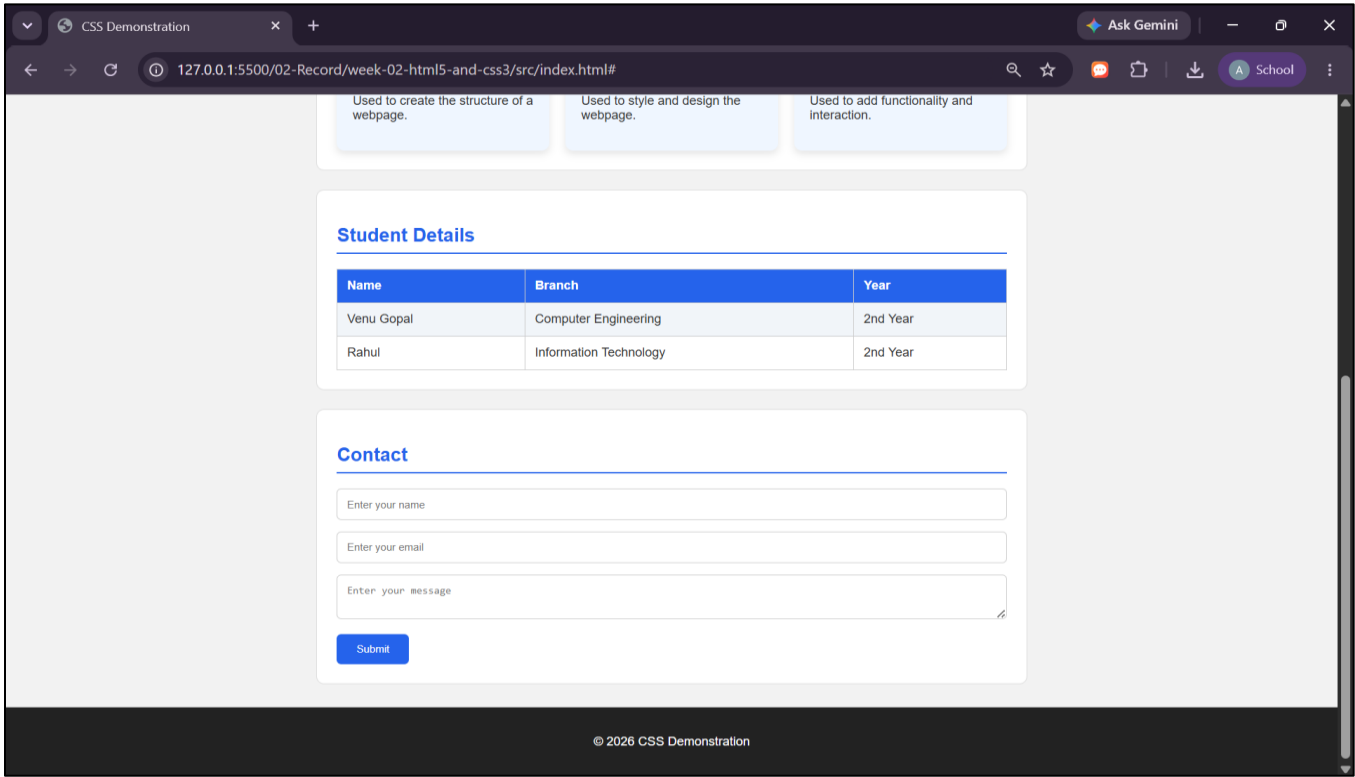
Screenshot 1: Main Webpage(without css)



Screenshot 2: CSS Effects / Form



(Student table)



11. RESULT

The webpage was successfully designed and styled using basic and medium CSS concepts with an external stylesheet.

12. CONCLUSION

The experiment demonstrates the use of CSS for webpage layout, appearance, and interaction through selectors, Flexbox, pseudo-classes, transitions, transforms, tables, and form styling.